No well-ordering is isomorphic to one of its proper initial segments

The Open Logic Project contributors

openlogic:set-theory/ordinals:local-snapshot-20261013; CC BY 4.0

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Open Logic Project unit file sources/olp-set-theory-ordinals/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the least-mismatch construction, both transport directions, and the closing contradiction with the named external citation appear verbatim in the span.

Editorial note. The label wellordnotinitial names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lem environment in the unit source.

Allowed standard prerequisites (editorial).

- Strict order from a well-order on a mismatch set. Same unit, \olla-bel{wo:strictorder} (source line 1250). At line 783: the set of points where a bijection f : A -> A_a disagrees has a least element b, since f is not the identity and A is well-ordered.
- Extensionality for strict linear orders. Cross-unit citation \olref[sfr][rel][ord]{prop:extensionality-strictlinearorders} invoked at line 786; the target unit (Open Logic Project, Sets/Functions/Relations: Relations) is not staged in this corpus. At 786-788: the equivalence of the strict lower sets of b and f(b) forces b = f(b), contradicting the choice of b.

Source reference. Original proposition prop-extensionality-strictlinearorders: sources/olp-set-theory-ordinals/pinned-original-source.tex, source lines 786–786. Named citation invoked at source line(s) 786: the author's marker \olref[sfr][rel][ord]{prop:extensionality-strictlinearorders} resolves inside the full Open Logic Project to another unit (not staged in this corpus); the cited fact is used as a standard result and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference wo-strictorder: sources/olp-set-theory-ordinals/pinned-original-source.tex, source lines 1250–1250. Prerequisite pointer invoked at proof line(s) 783; it names a standard fact of the same unit that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *If $\langle A, <\rangle$ is a well-ordering with $a \in A$, then $\langle A, <\rangle \neq \langle A_a, <_a\rangle$*

Proof. For reductio, suppose $f\colon A \to A_a$ is an isomorphism. Since f is a bijection and $A_a \subsetneq A$, using wo-strictorder let $b \in A$ be the $<$-least !!element of A such that $b \neq f(b)$. We'll show that $(\forall x \in A)(x < b \leftrightarrow x < f(b))$, from which it will follow by prop-extensionality-strictlinearorders that $b = f(b)$, completing the reductio.

Suppose $x < b$. So $x = f(x)$, by the choice of b. And $f(x) < f(b)$, as f is an isomorphism. So $x < f(b)$.

Suppose $x < f(b)$. So $f^{-1}(x) < b$, since f is an isomorphism, and so $f^{-1}(x) = x$ by the choice of b. So $x < b$. □